

Opeyemi Peter Ojajuni, PhD

Research Scientist | Human-Centered Evaluation of AI & Immersive Systems

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SUMMARY

Research scientist (PhD) with 20 publications and 125 citations, specializing in study design, measurement and evaluation of immersive and AI systems. Evaluated a VR instruction intervention for 987 students over four years (quasi-experimental, ASEE 2026), and in a 68-student quasi-experiment a digital-twin learning workflow raised pass rates from 63.6% to 87.0% ($p = .016$). Lead writer on 9 grant proposals (NASA, ONR, NSF, USDA, NNSA). Also builds the ML and LLM tooling behind the research, from federated learning pipelines to citation-grounded RAG assistants.

TECHNICAL SKILLS

Study design & measurement: quasi-experimental design, pre/post program evaluation (Kirkpatrick model), factor analysis, survey validation, effect sizes, chi-square

Statistics & ML: time series (ARIMA), scikit-learn, Random Forest, SVM, XGBoost, CNN, federated learning (Flower, PyTorch)

AI systems: RAG, LangChain, FAISS, OpenAI API, local LLMs (Ollama), human-in-the-loop agent design

Immersive research platforms: Unity, VR/CAVE, digital twins, LiDAR reality capture, GIS (OpenStreetMap, ArcGIS)

Tools: Python, SQL, Streamlit, Git/GitHub

RESEARCH EXPERIENCE

Postdoctoral Research Manager, SU XR Jags Research Laboratory

Feb 2024 – Present

Southern University and A&M College, Baton Rouge, LA

- Evaluated a VR/CAVE-based engineering-instruction intervention against traditional instruction for 987 students over four years (Fall 2021 – Spring 2025) using a quasi-experimental, between-subjects design (ASEE 2026).
- Co-developed a reality-capture → digital-twin → VR → 3D-print learning workflow; in a 68-student quasi-experiment, pass rates rose from 63.6% to 87.0% ($\chi^2(5) = 13.98$, $p = .016$) (journal manuscript, 2026).
- Evaluated a VR professional-development program (NTIA Connecting Minority Communities) for 88 teachers across two 2025 cohorts (pre/post surveys, 20-item self-efficacy battery, open responses; Kirkpatrick model); Fall cohort self-efficacy rose from 5.39 to 8.72 on a 1–10 scale (+62%), and 85% of Spring participants planned to implement VR that school year (self-reported, no control group).
- Validated a 29-item, 5-factor computational-thinking scale with engineering students using factor analysis (ASEE 2024), then measured medium effect sizes for algorithmic thinking and problem-solving under CAVE instruction (ASEE 2025).
- Built a mobility-aware federated intrusion-detection pipeline for connected vehicles (VeReMi data), benchmarked against FedAvg, FedProx, FedNova and FedBuff; reached 0.92 malicious recall and ROC AUC 0.79 on a frozen 199,663-sample test set with ~53% estimated communication savings, and documented the high false-positive rate as the blocker to deployment.
- Built, trained and validated deep-learning (CNN) and classical ML models (SVM, XGBoost, Random Forest, KNN) on multimodal data, including CNN work for the Louisiana Department of Health ICLERT project.

Graduate Research Assistant, Southern University and A&M College

Aug 2018 – Dec 2023

Baton Rouge, LA

- Built a Random Forest student-success classifier on the UCI Student Performance data and shipped it as a public Streamlit web app (SITE 2023).
- Designed a software-defined-networking framework that detects and mitigates DDoS attacks on IoT networks through API-based bandwidth policies (IJTD, 2020).
- Co-authored an ARIMA-based analysis of minority graduate trends in construction-management programs (ASEE 2023).

Data Scientist Intern, Amazon (Amazon Science – Mechatronics & Sustainable Packaging)

Jun 2023 – Sep 2023

Seattle, WA

- Ran the end-to-end analytics workflow for an automated-fulfillment initiative: problem framing, hypothesis design, modeling, validation, and leadership-facing analysis.
- Wrangled 1M+ logistics and transactional records with Python and SQL; built capacity-forecasting models with quantified uncertainty to inform capital-investment planning.

Earlier: Network & Security Engineer, Staunton and Lycett Ltd, Nigeria (Mar 2015 – Feb 2018)

SELECTED PUBLICATIONS (20 PUBLICATIONS, 125 CITATIONS; FIRST-AUTHOR ML PAPER CITED 92 TIMES)

- Ojajuni, O. P., Lawson, A. H. (2026). NSF IUOE: Evaluating academic performance in VR-enhanced engineering education: a quasi-experimental study. ASEE 2026.
- Ojajuni, O. P., Ismail, Y., Dawan, F., Lawson, A. H. (2025). Exploring the impact of knowledge acquisition in a CAVE on engineering students' computational thinking skill levels. ASEE 2025.
- Ojajuni, O. P., Dawan, F., Ismail, Y., Lawson, A. H. (2024). Immersive engineering learning and workforce development: pushing the boundaries of knowledge acquisition in a CAVE. ASEE 2024.
- Ojajuni, O., Ayeni, F., Akodu, O., et al. (2021). Predicting student academic performance using machine learning. ICCSA 2021, 481–491.
- Ojajuni, O., Ismail, Y., Lawson, A. (2020). Distributed denial-of-service attack detection and mitigation for the Internet of Things. International Journal of Technology Diffusion, 11(2), 18–32.
- Ojeh, A., Ojajuni, O., Warren, B., Street, D., Lawson, A., Stubblefield, M. (2025). A systematic review on the application of virtual reality in public health training. SITE 2025.

GRANTS AND PROPOSALS

- Lead writer on 9 federal, state and internal grant proposals (NASA, ONR, NSF, USDA, NNSA, LTRC, and a university research award), covering VR workforce training, protocol-grounded LLM decision support, and RAG-based advisory systems; largest single request \$1M (NASA).
- Co-PI on a 2026 research proposal for explainable, protocol-grounded LLM decision support in disaster preparedness, including a human-centred evaluation testbed built on immersive emergency simulations.

SELECTED PROJECTS

- Built InsightBot, a retrieval-augmented research assistant (LangChain, OpenAI embeddings, FAISS, Streamlit) that ingests article URLs and answers with source citations; public demo.
- Building Pathloom, a local-first career-intelligence app (Next.js, FastAPI, Ollama) in which an LLM agent proposes where documents belong and a human approves every change to stored facts; 33 automated test files.

EDUCATION

PhD, Science & Mathematics Education, Southern University and A&M College, 2023. Dissertation: Exploring the Impact of Immersive Technology on Engineering Students' Computational Thinking Skills

MBA (in progress), Southern University and A&M College

MSc Mobile & Satellite Communication, University of Surrey, UK, 2014 · BEng Computer Engineering, Covenant University, Nigeria, 2012

CERTIFICATIONS

AWS Certified AI Practitioner (active through Jul 2028) · IBM Artificial Intelligence Practitioner (2022)